Week #10: 2 Nov- 6 Nov

Subject/:الموضوع Chemistry

Unit Title: عنوان الوحدة Unit 4: Oxidation and reduction

Grade الصف: 10

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: Voltaic cells	**I can Understand and apply** the principles of a **voltaic (galvanic) cell** to explain how chemical energy is converted into electrical energy through redox reactions.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 4.2 Ensure access to tools and assistive technologies.	McGraw Hill unit 4, PPT; Chemistry Notebook; Teams; Toddle Lab report
Lesson 2 Voltaic cells lab	I can Experiment and explain how a voltaic (galvanic) cell works by calculating its potential difference using standard electrode potentials.		
Homework الواجبات	Review **redox reaction and oxidation numbers**.		
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Hello Parents, Please encourage your child to review and practice Science regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide basic *assessment information, topics, and expectations*; however, they are intended as a guide only and will not contain detailed study materials, these are found in the class teams. Effective preparation for graded formatives and summative requires students to actively engage with the learning materials provided in their Toddle class drive. Thank you for partnering with us to support your child's learning and success. Ms. Stacey (Science Coordinator)		

#الأسبوع Week 30

Technology Integration دمج التكنولوجيا	Use **digital reaction models/PPT animations** to visualize electron transfer and changes in oxidation numbers during redox reactions. Students may use digital periodic-table resources to support oxidation-number determination.
Cross-curricular links الروابط بين المواد الدراسية	**Physics: movement and transfer of electrons and electrical charge.** **Mathematics: applying positive and negative numbers and algebraic reasoning to determine unknown oxidation numbers.** **Environmental Science: oxidation processes such as corrosion and the role of redox chemistry in batteries and environmental systems.**
External Assessment Integration دمج التقييمات الخارجية	MAP , CAT 4
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	thinking Skills – Critical Thinking Analyze oxidation and reduction reactions, calculate cell potential (E°cell), and justify the choice of electrodes. Research Skills – Information Literacy Interpret standard electrode potential tables and use scientific data to predict cell voltage. Thinking Skills – Transfer Apply knowledge of redox reactions to explain how batteries produce electricity and connect this to sustainable energy applications in the UAE. Social Skills – Collaboration Work cooperatively to construct a voltaic cell, assign group roles, and compare experimental results. Communication Skills Explain electron flow, present calculations, and communicate experimental findings using correct

	scientific terminology. Self-Management Skills – Organization Follow laboratory procedures, handle materials safely, record observations, and complete calculations accurately.
Standards المعايير	**HS-PS1-2: Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms, trends**
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**How can we use evidence that we cannot directly see, such as electron transfer and changes in oxidation numbers, to explain what happens during a redox reaction?**
Success Criteria معايير النجاح **(Measurable)**	لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **L1:** 🟢 **MUST — DOK 1–2** **IDENTIFY the anode, cathode, salt bridge, and direction of electron flow in a voltaic cell.** 🟠 **SHOULD — DOK 3** **EXPLAIN how oxidation at the anode and reduction at the cathode produce electron flow and convert chemical energy into electrical energy.** 🔴 **COULD — DOK 4** **DESIGN a voltaic cell for a real-life application and JUSTIFY the choice of electrodes using scientific evidence.** **L2:** 🟢 **MUST — DOK 1–2** **IDENTIFY the standard reduction potentials of two electrodes from a given data table and STATE the formula for cell potential.** 🟠 **SHOULD — DOK 3** **CALCULATE the standard cell potential (E°cell) using electrode potentials and EXPLAIN how the result relates to electron flow.** 🔴 **COULD — DOK 4** **DESIGN and EVALUATE a voltaic cell investigation using different electrode materials, COMPARE calculated and measured voltages, and JUSTIFY which combination is most suitable.**

National Identity Domain مجالات الربط بالهوية الوطنية	Conservation الحفاظ ع لالبيئة Global Understandingه التفاهم العال Heritage ال رتاث **Conservation الحفاظ على البيئة | Global Understanding التفاهم العالمي** **UAE Values & Sustainable Development** Connect redox chemistry to **battery technology, corrosion prevention, and responsible battery disposal in the UAE**. Students discuss how understanding oxidation and reduction can support **safer energy use, protection of infrastructure, and environmental responsibility**. This also connects nicely with the **responsible battery disposal and environmental responsibility** point already included in the previous planner format